



Trader Behavior vs Market Sentiment (Fear & Greed Analysis)

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1. Executive Summary

This project analyzes the relationship between **market sentiment (Fear & Greed Index)** and **trader behavior** using historical trading data from Hyperliquid. The goal is to understand how sentiment affects **risk-taking, trading activity, and profitability**.

Key Findings:

- **Fear periods show the highest trading activity and the highest average risk per trade.**
- **Greed periods contain the largest extreme losses**, indicating overconfidence-driven blowups.
- **Neutral periods are significantly more stable** with fewer extreme outcomes.
- **Extreme Greed surprisingly shows lower risk per trade**, suggesting partial profit-taking or cautious behavior.
- Traders behave **more emotionally and aggressively during Fear** than during Greed.

Overall, **Fear is more dangerous than Greed in terms of systemic risk buildup**, while **Greed is where the largest crashes happen**.

2. Problem Statement

Financial markets are strongly influenced by human psychology. This study aims to:

- Analyze how **trader behavior changes under different sentiment regimes**
- Measure impact on:
 - Profit & Loss (PnL)

- Risk (trade size in USD)
 - Trading activity
 - Identify **hidden behavioral patterns** that can help design **better risk management systems**.
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3. Datasets Used

3.1 Fear & Greed Index Dataset

Columns:

- date
- classification (Extreme Fear, Fear, Neutral, Greed, Extreme Greed)

This dataset represents **daily market sentiment**.

3.2 Historical Trader Data (Hyperliquid)

Key columns used:

- Account
- Timestamp IST
- Execution Price
- Size USD
- Closed PnL
- Direction

This dataset contains **real executed trades** with profit/loss.

4. Data Processing & Feature Engineering

Steps performed:

- Converted timestamps to datetime format (DD-MM-YYYY handled properly)
- Extracted date from trade timestamps
- Merged trades with sentiment using date

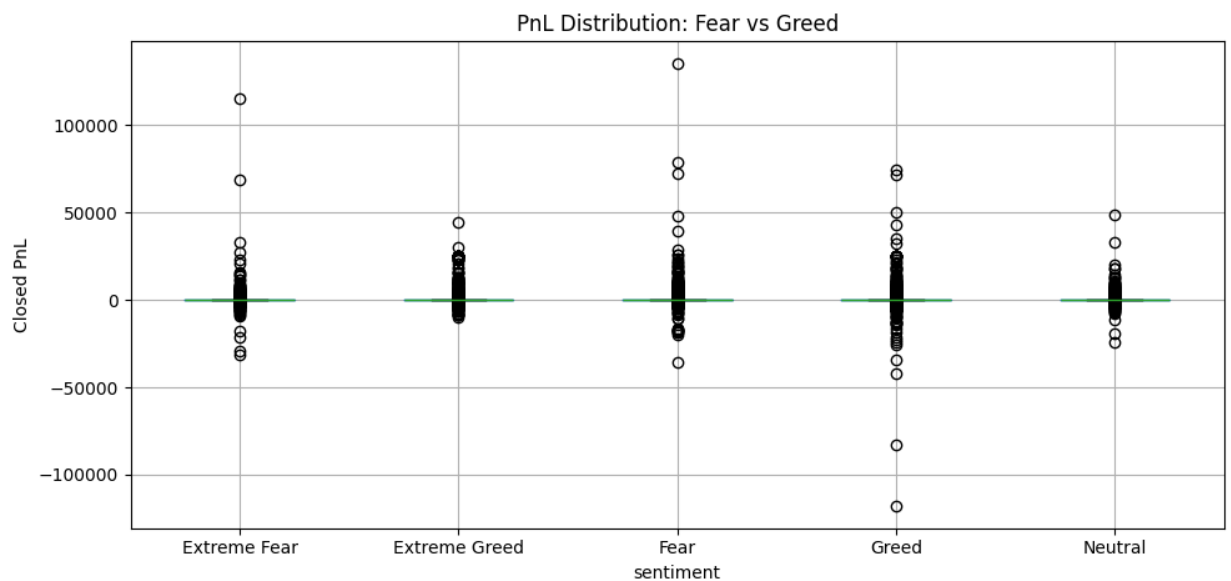
- Created new features:
 - is_profit → whether trade is profitable
 - abs_pnl → absolute PnL
 - risk → trade size in USD
 - trade_value → trade size in USD

Invalid or unmatched rows were removed.

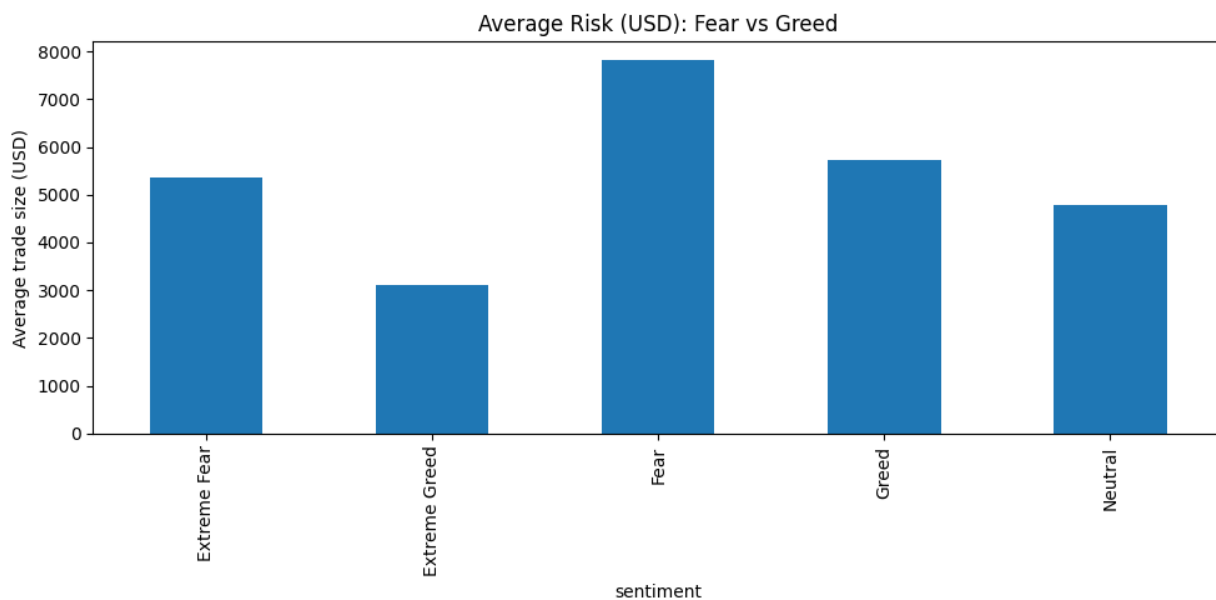
5. Exploratory Data Analysis

Three key analyses were performed:

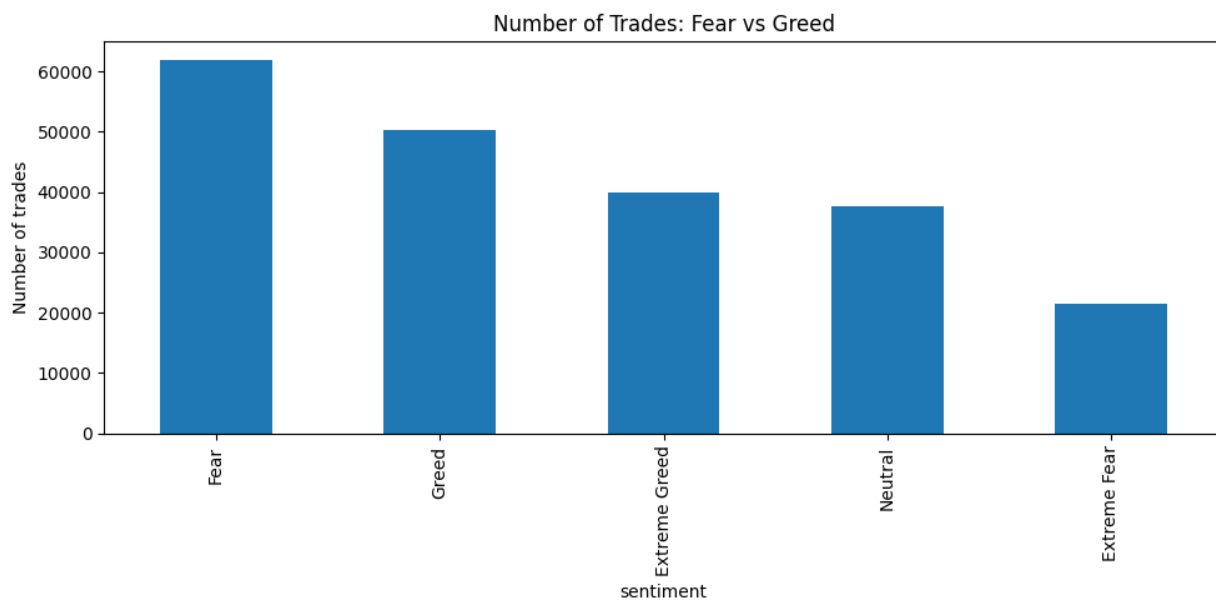
1. PnL Distribution vs Sentiment



2. Average Risk (Trade Size USD) vs Sentiment



3. Number of Trades vs Sentiment



(Charts attached in the outputs folder and embedded in this report.)

6. Key Findings & Interpretation

6.1 PnL Distribution: Fear vs Greed

Observations:

- All regimes show many small trades near zero PnL.
- **Fear and Greed show extreme outliers** (very large profits and very large losses).
- **Greed has the largest negative outliers** (catastrophic losses up to ~ -120k USD).
- **Neutral periods are much more stable** with fewer extreme outcomes.

Interpretation:

Greed periods produce the largest blow-ups due to overconfidence and excessive risk-taking. Fear periods are highly volatile with both big wins and big losses. Neutral regimes are the most stable and controlled.

6.2 Average Risk (USD) vs Sentiment

Observed ranking (highest to lowest risk):

1. Fear (Highest)
2. Greed
3. Extreme Fear
4. Neutral
5. Extreme Greed (Lowest)

Interpretation:

Traders take the **largest position sizes during Fear**, not Greed. This suggests panic trading, revenge trading, and aggressive averaging-down behavior.

Surprisingly, during Extreme Greed, traders reduce position sizes, possibly due to profit booking or increased caution after strong rallies.

Fear is the most dangerous regime in terms of risk exposure.

6.3 Number of Trades vs Sentiment

Observations:

- **Fear has the highest trading activity**
- Greed is second highest
- Extreme Fear has the lowest activity

Interpretation:

Fear triggers panic, forced liquidations, and emotional overtrading. Traders act more during Fear than Greed.

Extreme Fear shows reduced participation, possibly due to traders staying out or liquidity drying up.

7. Combined Behavioral Insight (Core Conclusion)

Fear and Greed cause **different types of failure modes**:

- **Fear:**
 - Overtrading
 - Oversized positions
 - Panic decisions
 - High systemic risk buildup
- **Greed:**
 - Overconfidence
 - Holding losers too long
 - Rare but catastrophic losses

Neutral sentiment periods are the healthiest for trading behavior.

8. Business Recommendations

Based on this analysis:

1.  Implement **sentiment-aware risk limits**
 - Reduce position size caps during Fear
 2.  Enforce **stricter loss controls during Greed**
 3.  Add **trade frequency throttling during Fear**
 4.  Encourage **capital preservation mode** during extreme sentiment regimes
 5.  Use sentiment as a **first-class input to risk management systems**
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9. Conclusion

This study demonstrates that:

- **Fear is more dangerous than Greed from a systemic risk perspective**
- **Greed is where the biggest blowups happen**
- Market sentiment is a **powerful predictor of trader behavior and risk**

Incorporating sentiment-aware controls can significantly improve **risk-adjusted performance and platform stability**.

10. Appendix

- Full analysis code in: notebook_1.ipynb
- Charts in: outputs/
- Aggregated tables in: csv_files/sentiment_summary.csv